

JINNAH INTL, Pakistan

WMO: 417800

Lat: 24.907N

Lon: 67.161E

Elev: 31

StdP: 100.96

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	11.9	-9.8	1.6	22.5	-6.1	2.3	22.2	10.1	20.4	8.7	21.0	1.9	30	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.2	39.0	22.7	37.2	23.1	36.0	23.5	28.2	33.4	27.9	32.8	27.5	32.3	4.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	23.0	31.1	26.7	22.4	30.9	26.2	21.7	30.6	91.9	33.4	90.0	32.3	88.3	32.6	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.7	8.5	7.7	DB	7.5	42.3	2.4	1.3	5.8	43.3	4.4	44.0	3.1	44.8	1.3	45.8
			WB	2.7	30.0	1.8	0.8	1.4	30.6	0.3	31.1	-0.7	31.5	-2.0	32.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.2	19.4	22.0	26.3	29.6	31.5	32.1	30.8	29.6	29.9	29.4	25.2	20.8
	DBStd	4.59	2.07	2.55	2.37	1.69	1.23	1.30	1.25	1.15	1.40	1.69	2.17	2.23
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	21	13	3	0	0	0	0	0	0	0	0	0	4
	CDD10.0	6297	291	336	506	588	667	664	646	609	597	602	456	336
	CDD18.3	3276	46	106	248	338	408	414	387	351	347	344	206	82
	CDH23.3	40489	379	887	2492	4141	5616	5942	5213	4341	4296	4273	2155	754
CDH26.7	20130	42	261	1131	2071	3151	3546	2738	1878	1944	2213	973	183	
Wind	WSAvg	3.6	2.5	2.8	3.1	3.8	4.9	5.2	5.3	4.8	4.2	2.6	2.1	2.3
Precipitation	PrecAvg	163	8	7	3	1	0	14	54	45	20	3	1	7
	PrecMax	472	86	33	32	30	5	110	272	241	212	37	8	62
	PrecMin	13	0	0	0	0	0	0	2	0	0	0	0	1
	PrecStd	129	16	10	7	6	1	29	76	61	45	8	2	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	33.0	38.0	40.2	41.0	40.6	36.5	36.0	39.0	39.8	35.5	31.6
		MCWB	16.4	18.3	20.2	19.6	21.1	25.9	27.3	26.7	24.6	21.3	19.0	17.7
	2%	DB	27.9	31.0	35.7	37.8	38.1	37.4	35.0	33.8	35.9	37.8	34.1	30.0
		MCWB	15.9	18.1	19.3	20.6	23.4	26.6	27.2	26.5	24.5	21.1	19.1	17.5
	5%	DB	26.9	29.8	33.9	35.9	36.1	35.9	34.0	32.8	34.0	36.1	33.0	28.8
		MCWB	16.0	17.4	19.3	21.6	25.2	27.0	26.8	25.9	25.5	21.0	18.9	16.9
	10%	DB	25.8	28.1	32.1	34.2	34.9	34.9	33.1	31.9	32.9	34.8	31.8	27.2
		MCWB	15.3	17.0	19.6	22.5	25.9	26.9	26.5	25.6	25.2	21.5	19.0	16.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	22.6	24.5	27.0	28.6	29.1	28.6	28.1	28.1	26.9	24.9	22.5
		MCDB	25.5	28.1	30.8	33.3	34.6	35.3	33.1	32.4	32.8	32.2	29.6	27.4
	2%	WB	19.7	21.7	23.6	26.1	27.8	28.4	28.0	27.3	27.1	26.2	23.9	21.2
		MCDB	24.1	26.4	29.4	32.2	33.2	33.9	32.5	31.2	31.5	31.5	28.9	26.1
	5%	WB	18.7	20.9	22.9	25.5	27.3	28.0	27.6	26.9	26.6	25.7	22.9	20.0
		MCDB	23.3	25.4	28.4	31.1	32.5	33.2	32.0	30.6	31.0	30.9	28.1	25.0
	10%	WB	17.5	20.1	22.2	25.0	26.9	27.6	27.2	26.5	26.1	25.2	22.0	18.7
		MCDB	22.9	24.6	27.8	30.2	32.0	32.7	31.4	30.2	30.7	30.1	27.2	24.2
Mean Daily Temperature Range	5% DB	MDBR	12.8	12.6	11.9	10.4	7.8	6.2	4.8	4.8	6.6	11.5	13.6	13.5
		MCDBR	13.7	14.1	14.2	13.1	10.4	8.0	6.0	6.2	8.9	14.0	14.4	14.7
		MCWBR	6.8	5.9	5.3	4.9	4.3	3.3	2.3	2.4	3.1	5.2	6.0	7.2
	5% WB	MCDBR	11.8	10.6	10.7	9.2	7.4	6.6	5.4	5.3	7.1	9.4	12.1	13.0
		MCWBR	7.1	5.0	4.6	3.8	3.2	2.9	2.5	2.5	2.9	3.8	5.6	7.4
Clear-Sky Solar Irradiance	taub	0.446	0.458	0.513	0.566	0.635	0.733	0.793	0.727	0.621	0.539	0.515	0.486	
	taud	2.040	2.041	1.902	1.817	1.715	1.565	1.506	1.605	1.769	1.868	1.881	1.933	
	Ebn at Noon	792	817	793	760	706	636	598	637	695	731	714	727	
	Edh at Noon	151	162	195	217	240	277	293	265	220	190	176	163	
All-Sky Solar Radiation	RadAvg	4.39	5.39	6.45	7.18	7.41	6.81	5.52	5.40	5.91	5.66	4.66	4.16	
	RadStd	0.18	0.18	0.18	0.18	0.16	0.32	0.41	0.29	0.32	0.20	0.17	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	-0.48	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page